

# VISHNU BHARDWAJ

B.Tech Computer Science Engineering Student

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## PROFESSIONAL SUMMARY

Backend-focused software developer with hands-on experience building production-grade REST APIs, distributed systems, authentication services, AI-integrated applications, and scalable web platforms. Proficient in Python, Java, and JavaScript, with experience in MongoDB, Redis, microservices, deployment, server administration, and production infrastructure. Built and deployed systems handling approximately 1,000 requests per hour and contributed to a reported 20% growth improvement. Seeking Backend Developer or Software Developer internship opportunities.

## TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL

DSA & Problem Solving: Data Structures and Algorithms in Java and Python

Backend Development: Node.js, Express.js, FastAPI, REST APIs, WebSockets, Authentication & Authorization, JWT, Microservices, API Gateway Patterns, RBAC

Databases & Caching: MongoDB, PostgreSQL, MySQL, Redis, Redis Queues

Distributed Systems: RabbitMQ, Redis-based Message Queues, Microservices Architecture, System Design Patterns

AI & ML: RAG Architecture, ChromaDB, Embeddings, Ollama, MFCC, Spectrograms, Keyword Spotting, TTS Dataset Generation

Cloud & DevOps: AWS EC2/S3, Render, Hostinger, Docker, Linux/Ubuntu, SSH, Server Deployment, DNS, Domain Configuration, Reverse Proxy

Development Tools: Git, GitHub, VS Code, npm, Python Virtual Environments, PyPI Package Publishing

## PROJECTS

### Scalable E-commerce Platform — Production Application

Scalable Web Application, Deployment, Production Infrastructure

- Designed and deployed a scalable e-commerce application capable of managing approximately 1,000 requests per hour.
- Improved production application performance and business growth, contributing to a reported 20% increase in growth.
- Handled deployment, domain configuration, hosting, and production environment management across modern hosting infrastructure.

### ShipBihar — Logistics & Courier Aggregation Backend

Node.js, Express.js, MongoDB, Redis, RabbitMQ, JWT, Razorpay API

- Implemented JWT-based authentication with refresh token rotation, OTP verification, and email validation for secure user onboarding.
- Built wallet system with transaction history and integrated Razorpay for UPI payment processing and payment reconciliation.
- Designed Redis-cached pricing logic and AWB generation service for courier rate aggregation across multiple providers.
- Architected RabbitMQ-based asynchronous shipment processing pipeline for reliable order handling and status notifications.
- Deployed backend on Vercel and Render with MongoDB Atlas integration; managed DNS and domain configuration for production.

### FastAPI Authentication Microservice

FastAPI, Pydantic, MongoDB, Beanie, Motor, JWT, Email Service

- Built production-ready authentication service with JWT access/refresh token systems and automatic refresh token rotation.
- Implemented OTP-based email verification, password reset flow, and session management with token blacklisting.
- Designed Role-Based Access Control (RBAC) for admin/user permission management.
- Structured the codebase using repository and service architecture patterns for maintainability and testability.
- Configured async MongoDB queries with Beanie ODM and Motor drivers for non-blocking database operations.

## **Smart Queue Management System — Hospital/Store Queue Backend**

Express.js, MongoDB, Redis, JWT, Node.js

- Designed Redis atomic counter system for daily token generation and real-time patient position tracking using INCR operations.
- Implemented patient queue management with next-patient logic and role-based access for doctors, receptionists, patients, and admins.
- Built notification system to alert patients of position changes and wait times.
- Architected JWT-secured API endpoints for queue operations, token generation, and patient status retrieval.

## **Personal Deployment Platform**

Node.js, Express.js, Ubuntu Linux, SSH, AWS EC2, Docker

- Built Node.js deployment server capable of cloning repositories, installing dependencies, and provisioning application instances.
- Implemented Linux process management and port assignment for concurrent application deployment.
- Configured Ubuntu server with SSH key authentication, networking concepts, and reverse proxy setup.
- Deployed personal infrastructure on AWS EC2 with domain configuration and DNS management.
- Demonstrated end-to-end understanding of how applications transition from development to production server infrastructure.

## PROJECTS — CONTINUED

### ORBII — AI Voice Keyword Detection System

Python, FastAPI, Audio Processing, Machine Learning, GPU Training

- Built custom speech dataset using Piper TTS and audio augmentation (pitch shifting, noise injection) with MFCC and spectrogram features.
- Trained keyword spotting models for SOS emergency detection with low-latency inference optimized for edge devices.
- Implemented real-time voice detection pipeline with FastAPI backend for inference and result streaming.
- Leveraged GPU-based training in Google Colab for efficient model optimization.

### Kids Garden — School CMS Backend

FastAPI, PostgreSQL, Supabase, JWT, JSONB Data Types

- Developed FastAPI CRUD APIs for managing school content including teachers, facilities, admissions, and testimonials.
- Implemented JWT-based admin authentication and role-based content management.
- Used PostgreSQL JSONB for flexible schema design of gallery and testimonial data structures.
- Integrated Supabase Storage for image uploads and public file serving.

### GitHub Code RAG System

FastAPI, Ollama, ChromaDB, Vector Embeddings, Qwen2.5-Coder

- Built Retrieval-Augmented Generation (RAG) system for querying GitHub repositories using local LLMs.
- Implemented document chunking pipeline and vector embeddings using ChromaDB with 17,000+ code chunks.
- Designed FastAPI endpoints for repository indexing, semantic search, and context-aware code retrieval.
- Leveraged Ollama and Qwen2.5-Coder for local inference without external API dependencies.

## EDUCATION

B.Tech Computer Science Engineering (Pursuing)

Secondary School: 92% | Higher Secondary: 92.6%

## CERTIFICATIONS & OPEN SOURCE

- Published PyPI Package: `mongo_agent_mcp` — MCP server for MongoDB integration with AI agents
- GitHub: Active contributor with repositories demonstrating full-stack implementation, deployment experience, and system design